MeterTrace

Electricity Theft Detection and
Consumption Anomaly Analysis

Project overview

MeterTrace investigates whether daily smart-meter readings can help identify customers who may need an electricity-theft inspection. It combines a supervised sequence classifier with an unsupervised anomaly detector and compares both with traditional machine-learning baselines.

The aim is to support inspection prioritisation, not to declare that a customer has committed theft. Every flag should be treated as a lead for human review.

Dataset	SGCC smart-meter data
Customers	42,372 customer records
Labels	3,615 theft-labelled (8.5%); 38,757 normal-labelled
Methods	Logistic Regression, Random Forest, Isolation Forest, LSTM and dense autoencoder
Reported progress	Scope and dataset profiling completed; initial EDA cells written and tested. Model results remain pending.

Current stage

The project is at the data-understanding stage. Preprocessing, model training, evaluation and the interactive investigation demo are planned next steps.

1. Problem and dataset

Electricity theft can include meter tampering, bypassed meters and illegal connections. Since physical inspection of every customer is costly, utilities can benefit from a ranked list of accounts whose readings warrant review.

The project uses the SGCC smart-meter dataset associated with Zheng et al., IEEE Transactions on Industrial Informatics (2018). The initial profile identified data-quality issues that must be addressed before modelling.

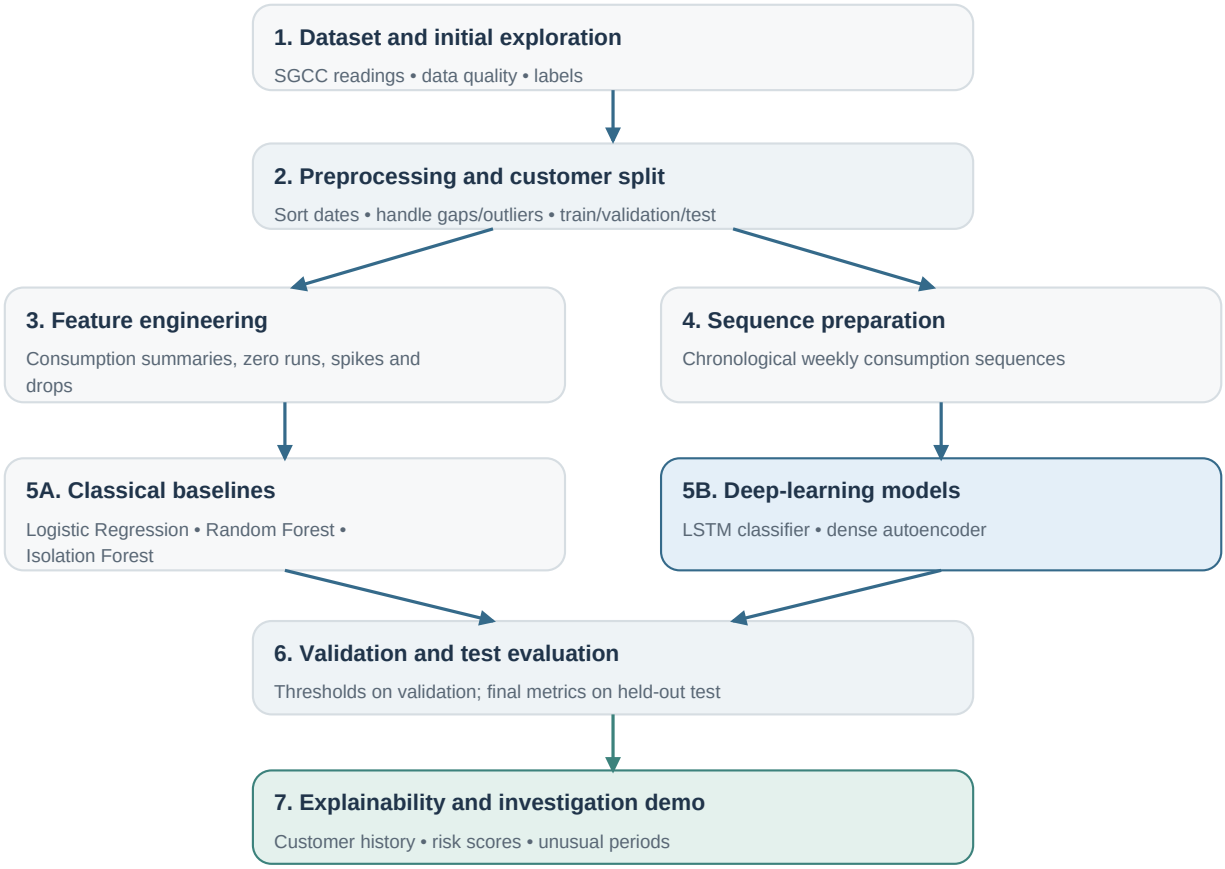
Profile item	Observation	Implication
Missing readings	25.6% overall; 31.5% for theft-labelled customers and 25.1% for normal customers	Document and test imputation choices.
Extreme readings	Values reach 800,003 kWh/day	Inspect and treat outliers consistently.
System-wide glitch	Median rises from 4.5 to 144 kWh on 3 Oct 2014	Treat the identified glitch as missing, per plan.
Class imbalance	Theft labels are 8.5% of customers	Accuracy alone is misleading.
Simple baseline	Average consumption gives ROC-AUC 0.66	More complex models should be compared against it.

Date columns must be sorted chronologically before sequence modelling. The initial analysis also found no strong basis for assuming theft customers have a weaker weekly rhythm; the models must learn useful distinctions from data.

2. Workflow diagram

The workflow separates data preparation, feature construction, model training, evaluation and the final customer investigation view. The diagram describes the planned sequence of work, not completed model results.

MeterTrace workflow

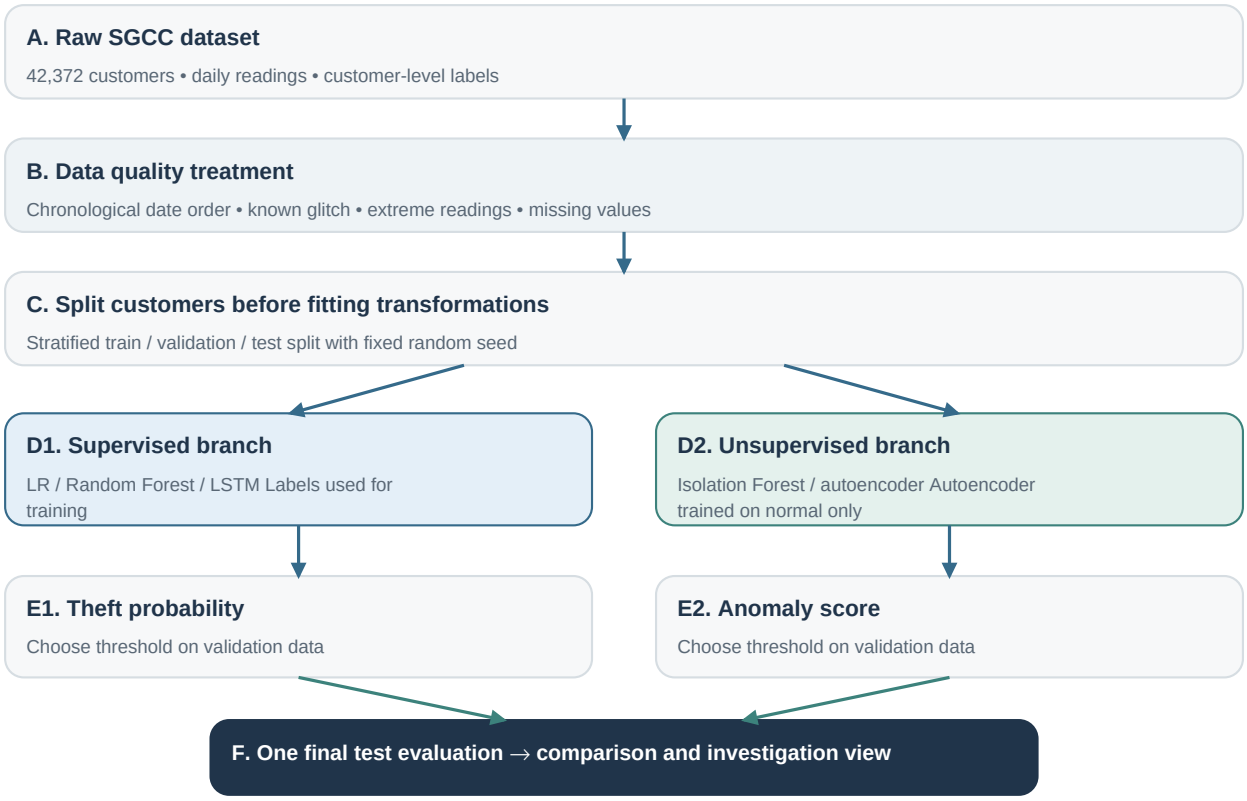


The classical and deep-learning models will be compared using the same held-out customers. Preprocessing transformations and decision thresholds must not be fitted using test data.

3. Model pipeline diagram

The pipeline below highlights the two branches and the controls used to avoid leakage. Supervised models use theft labels during training. The autoencoder learns only from normal customers in the training split; labels are used later to evaluate its scores, not to train it.

Model pipeline and evaluation controls



Model roles

Model	Role
Logistic Regression	Simple supervised baseline on engineered features.
Random Forest	Non-linear supervised baseline with feature-importance support.
LSTM classifier	Learns from weekly-aggregated consumption sequences and theft labels.
Isolation Forest	Classical unsupervised anomaly-detection baseline.
Dense autoencoder	Learns normal usage patterns; reconstruction error is the anomaly score.

4. Evaluation plan

Because 91.5% of customers are labelled normal, predicting every customer as normal would already achieve about 91.5% accuracy. The project will therefore report a broader set of metrics.

Metric	What it tells us
Precision	How many flagged customers are labelled theft.
Recall	How many theft-labelled customers are detected.
F1-score	Balance between precision and recall.
ROC-AUC / PR-AUC	Ranking quality; PR-AUC is especially informative for imbalanced labels.
False-positive rate	How often normal customers are incorrectly flagged.
Top-ranked precision	Usefulness of the highest-priority inspection list.

Evaluation safeguards

- Split customers before fitting imputers or scalers.
- Select probability and anomaly thresholds on validation customers.
- Evaluate the held-out test set once at the end.
- Use theft labels to evaluate unsupervised scores, never to train the unsupervised models.

5. Planned implementation stages

Step	Stage	Planned work
01	**EDA**	Missingness, zeros, outliers, class imbalance and consumption trends.
02	**Preprocessing**	Sort dates, handle the known glitch, cap outliers, impute gaps and split customers.
03	**Feature engineering**	Mean, spread, zero-day ratio, zero runs, spikes, drops, volatility and baseline changes.
04	**Baselines**	Train Logistic Regression, Random Forest and Isolation Forest.
05	**Deep learning**	Train LSTM and a dense autoencoder on normal training customers only.
06	**Evaluation**	Tune thresholds on validation data and score the held-out test set.
07	**Explainability**	Inspect feature importance, SHAP, reconstruction errors and customer histories.
08	**Interactive demo**	Review a customer's history, scores and unusual periods.

6. Expected use, limitations and next steps

If the models perform well, MeterTrace could help utilities rank accounts for review and make inspection planning more systematic. The final view is intended to show the evidence behind a flag, rather than provide an unexplained yes/no decision.

Limitations

• The labels are customer-level and do not identify when theft occurred; the label-assignment process is not documented in the source brief.
• The data comes from one utility and one region, so results may not transfer to other networks.
• Missing readings, extreme values and the known system-wide glitch may affect performance.
• Unusual consumption can result from legitimate behaviour or meter faults, not only theft.

Reported progress

The project scope has been defined, the dataset has been obtained and profiled, and initial EDA notebook cells have been written and tested. Model training, final evaluation and the interactive demo are still planned work.

Immediate next steps

1. Complete EDA and document preprocessing decisions.
2. Create leakage-safe customer splits and engineered features.
3. Train and benchmark classical models.
4. Train the LSTM and autoencoder.
5. Compare performance, inspect false positives and explain model outputs.
6. Build the investigation demo and document limitations.

Possible extensions

Explore a CNN using weeks-by-days inputs, LSTM autoencoders or attention models; validate on another public dataset; calibrate model outputs before combining scores; and investigate weather or holiday features.

> **Important:** A high theft probability or anomaly score is a lead for human investigation, not proof of theft.

Dataset reference

SGCC smart-meter dataset, released with Zheng et al., IEEE Transactions on Industrial Informatics (2018), as identified in the project brief. Add the full bibliographic citation in the final submission using the required citation style.